Protein Engineering

Module: SC3282

By

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Lecture 02:

- Amino acids and their properties.
- Protein structures and bonds.

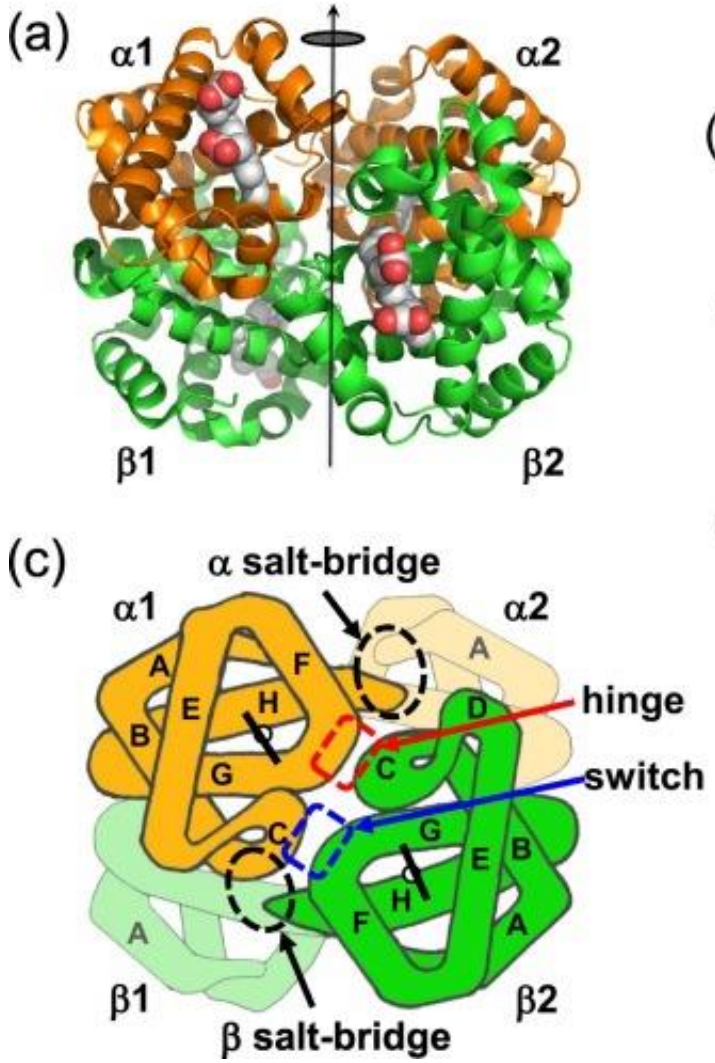
What is meant by the conformation of a protein?

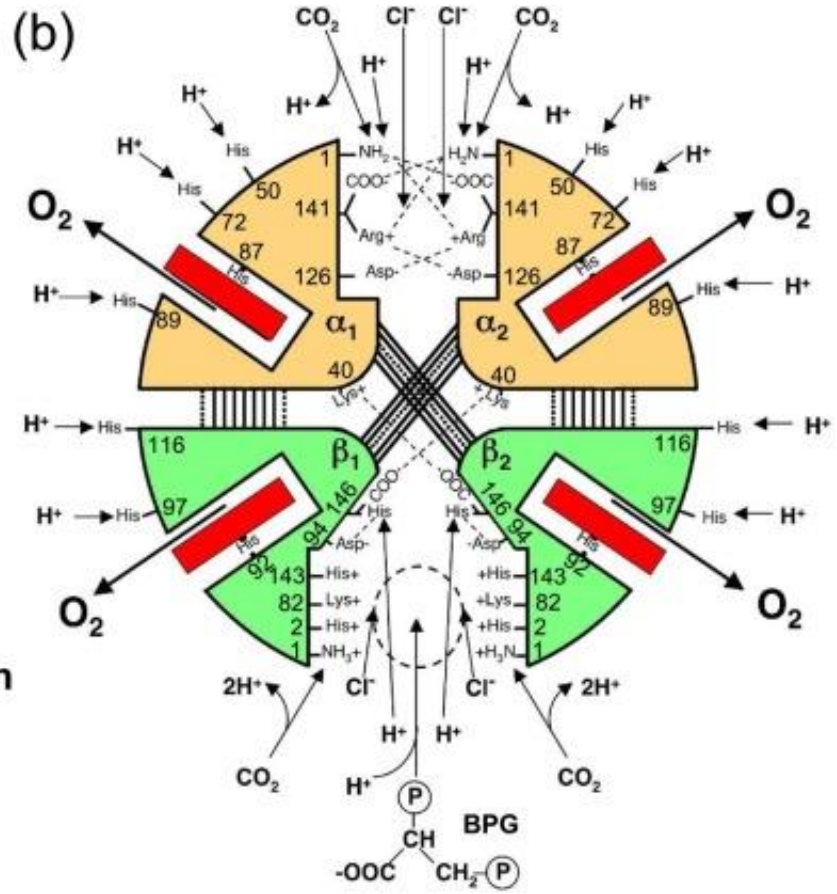
The characteristic of 3-dimensional shape of a protein, including the secondary, super-secondary (motifs), tertiary (domains) and quaternary structure of the peptide chain.

Why is a protein conformation important?

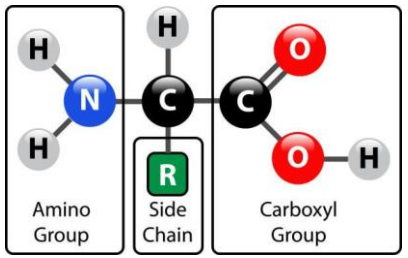
Protein conformation is of paramount importance in understanding biomolecular interactions. For protein like haemoglobin that shows allosteric behaviour, the binding of small molecules at a region of the protein affects its binding affinity with other molecules at a distant region.

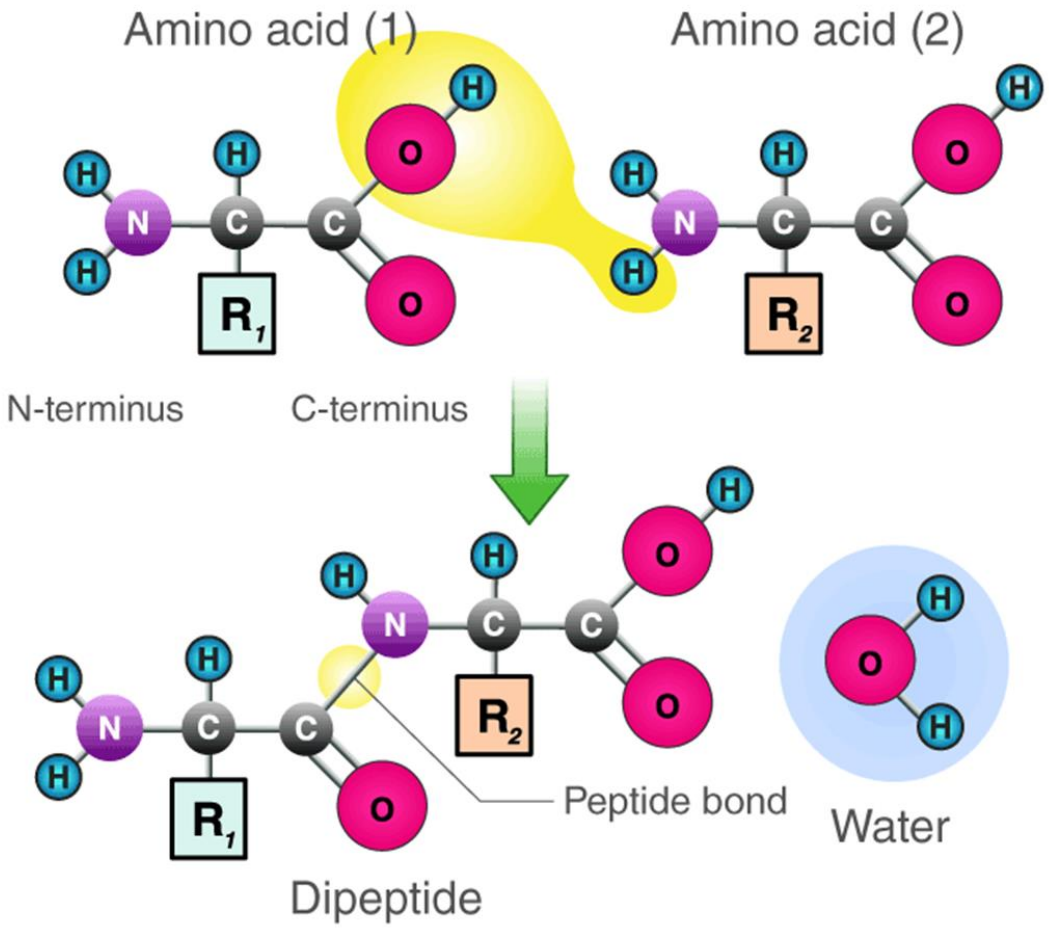
Allosteric transitions in hemoglobin



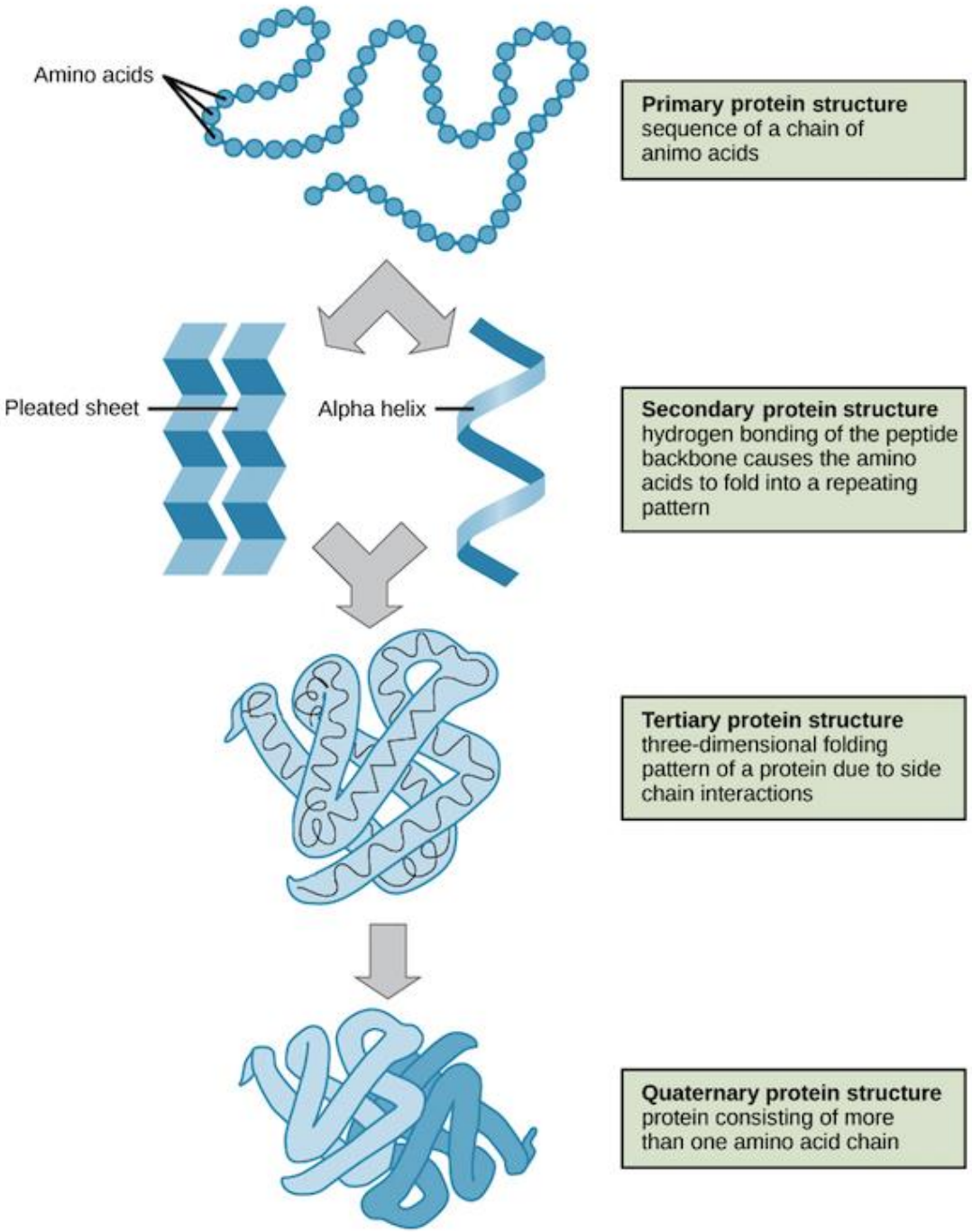


Amino Acids/ peptide bond formation



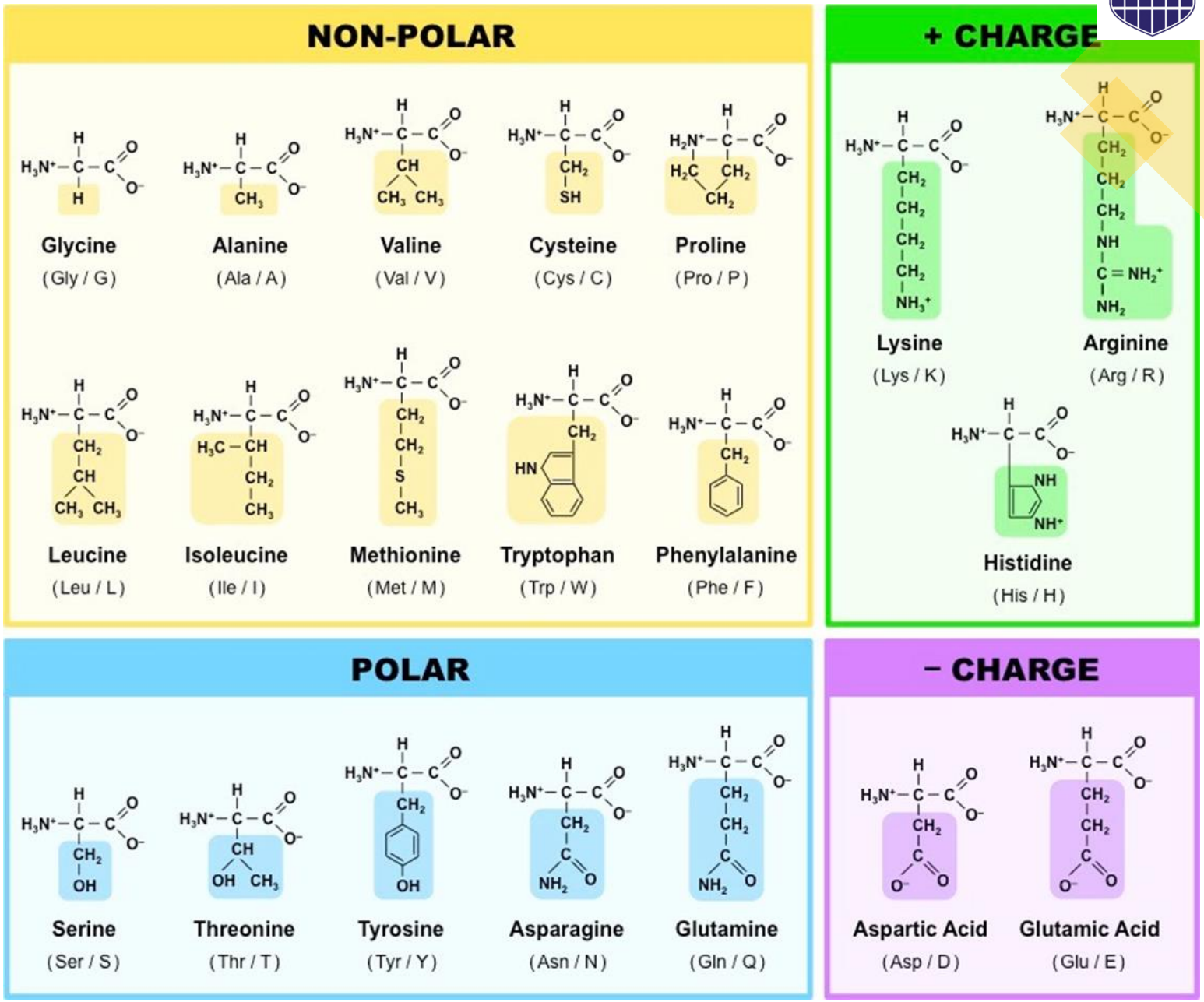


- Amino Acids
- Type of Amino Acids
- Primary Structure
- Secondary Structure
- Tertiary Interactions
- Quaternary structure



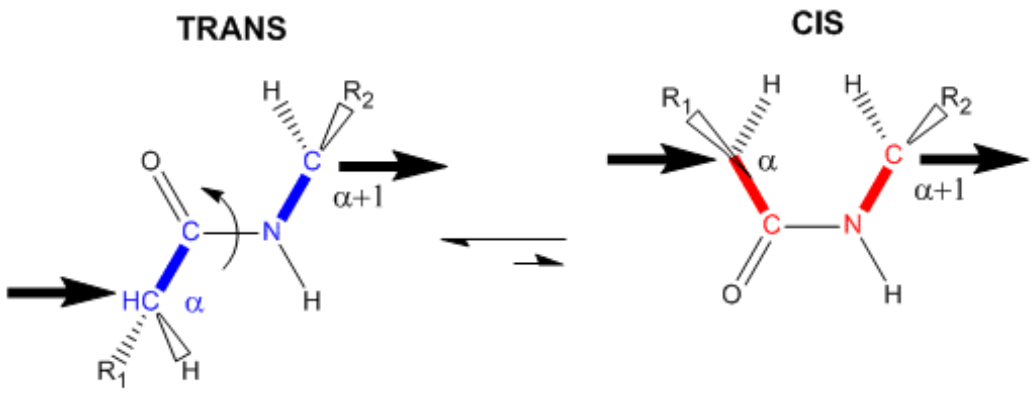
20 amino acids

- Since the amino acids (except glycine) have 4 different groups attached to the a-carbon, they are optically active stereoisomers.
- R-side chain-represents "a chemical group" and gives each amino acid its distinctive properties.
- There are only 20 different amino acid. These sidechains have a widespread range of chemical (e.g., charge etc) and structural (e.g. shape etc) diversity.

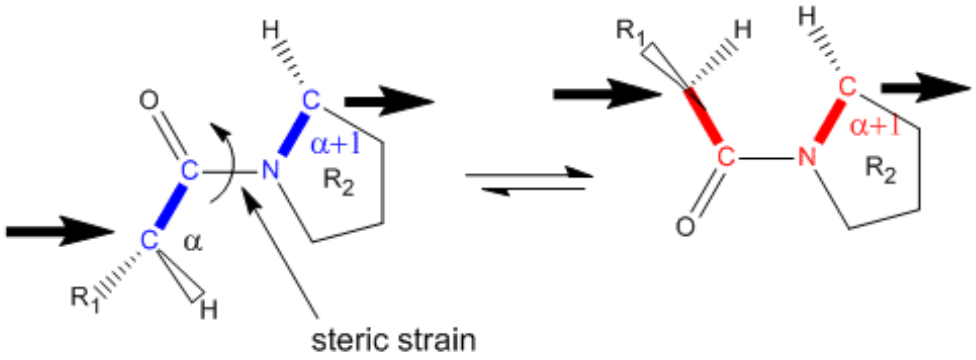


PEPTIDE BONDS: CIS OR TRANS

X-Y



X-Pro



Essential	Conditionally Non-Essential	Non-Essential
Histidine	Arginine	Alanine
Isoleucine	Asparagine	Asparatate
Leucine	Glutamine	Cysteine
Methionine	Glycine	Glutamate
Phenylalanine	Proline	
Threonine	Serine	
Tryptophan	Tyrosine	
Valine		
Lysine		

Polar and non-polar:

- A nonpolar molecule has no separation of charge, so no positive or negative poles are formed. In other words, the electrical charges of nonpolar molecules are evenly distributed across the molecule.

- Nonpolar molecules tend to dissolve well in nonpolar solvents, which are frequently organic solvents.

- All polar amino acids have either an OH or NH2 group (when in aqueous environment) and can therefore make hydrogen bonds with other suitable groups.

- Polar amino acid residues have a tendency to be on the outside of a protein, due to the hydrophilic properties of the side chain.

ALIPHATIC AMINO ACIDS:

Aliphatic amino acids are hydrophobic. They do not like to be in contact with water molecules in an aqueous solution. They are often located in the core of the protein, surrounded by the rest of the protein, and "shielded" by them from the aqueous surroundings.

- **Glycine (Gly; G)**: is the simplest and smallest of all amino acids, and the only one which is not optically active since it has a single hydrogen atom as it's sidechain.
- **Alanine (Ala; A)**: has a methyl group as it's sidechain.
- **Valine (Val; V)**: has a slightly longer sidechain again, and this time there is a branch. As the aliphatic side chains get longer, they are also more hydrophobic.
- **Leucine (Leu; L)**: similar to Valine except it has another methyl group.
- **Isoleucine (Ile; I)**: is again similar to leucine and valine except that the orientation of the atoms in the sidechain is slightly different. Isoleucine also has two centers of asymmetry.
- **Proline (Pro; P)**: the sidechain is bonded to both the a-carbon but also to the amino group.

AROMATIC AMINO ACIDS:

These are the amino acids which contain an aromatic ring as part of their sidechains. Because of the hydrophobic nature of aromatic rings these amino acids are highly hydrophobic.

- **Phenylalanine (Phe; F)** is the first of all the aromatic amino acids. It contains a phenyl ring attached to a methylene group. Due to the phenyl ring it is hydrophobic amino acid.

- **Tyrosine (Tyr; Y)** contains a hydroxyl group at the end of the phenyl ring. This makes tyrosine less hydrophobic than phenylalanine. It is also a reactive group, whereas the sidechains so far have all been unreactive.

- **Tryptophan (Trp; W)** has a slightly different ring attached to the methylene group. This is an indole ring and it is highly hydrophobic.

SULPHUR CONTAINING AMINO ACIDS:

There are two amino acids which contain a sulphur atom. They are both very special but for completely different reasons.

- **Cysteine (Cys; C)** contains a sulphydryl group (-SH). This is extremely reactive and can form hydrogen bonds. Cysteine is very important because it can also form di-sulphide bridges. Even though the -SH group of cysteine form hydrogen bonds the long aliphatic part of the side chain makes it quite hydrophobic.

- **Methionine (Met; M)** is a very special amino acid in that it is the "start“ amino acid in the process of translation (protein synthesis), and therefore, begins every single protein made. It too has a sulphur atom, but this time it is in a thioether linkage, and is relatively unreactive. Methionine has a highly hydrophobic sidechain.

HYDROPHILLIC AMINO ACIDS

These amino acids can be split into a several groups.

- Neutral
- Acidic
- Basic

ACIDIC AMINO ACIDS

These amino acids are highly polar, and are nearly always negatively charged at physiological pH

- **Aspartate (Asp; D)** is really aspartic acid. It is called aspartate because it is usually negatively charged at physiological pH and so it is named for the carboxylate anion. (Compare acetic acid and acetate.).

- **Glutamate (Glu; E)** is also called glutamic acid. The side chain of glutamate has a carboxylate group which has a negative charge at physiological pH.

NEUTRAL POLAR AMINO ACIDS

These amino acids are not charged at physiological pH. However, they all have groups on their side-chains which are polar and can form hydrogen bonds. For this reason, the amino acids are classed as hydrophillic.

- **Serine (Ser; S)** contains an aliphatic chain with a hydroxyl group. It looks like the hydroxylated version of alanine. The OH group make the amino acid highly reactive and hydrophillic as it readily forms hydrogen bonds.

- **Threonine (Thr; T)** is another neutral amino acid which has a highly reactive (and highly hydrophillic) hydroxyl group. This is an odd amino acid in that it contains two centers of asymmetry (two asymetric carbon atoms). This property is shared only by isoleucine.

- **Asparagine (Asn; N)** is the amide derivative of Aspartic acid. When the carboxylate sidechain is amidated the resulting amide is uncharged. There is a terminal amide group as opposed to the carboxyl group on aspartate.

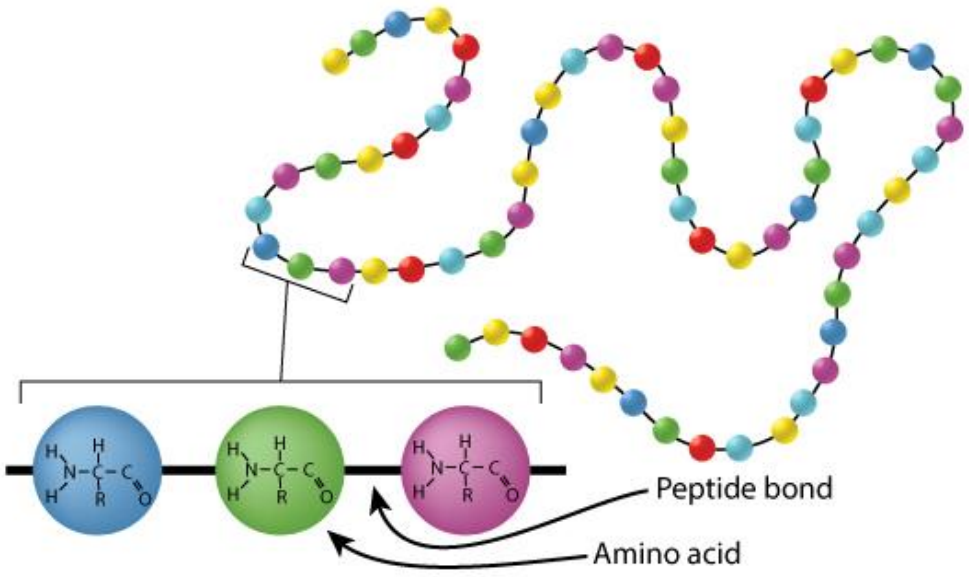
- **Glutamine (Gln; Q)** is similar to Asparagine, with a terminal amide instead of a carboxyl as in glutamate. These two are called the amide derivatives of their parent amino acids.

Primary Structure

- A protein has a completely defined order of amino acids, called its sequence.

- This defined sequence is called the primary structure of the protein. The sequence, or primary structure is specified by the sequence of the piece of DNA containing a gene for that protein, and the sequence is unique to the individual protein.

- All the information necessary to make a protein is contained in the primary structure. However, as we shall see soon, the primary structure is only the first level of structural complexity of proteins.



Secondary structure:

- Generally, proteins have a core of hydrophobic residues surrounded by a shell of hydrophilic residues.
- Since the peptide bonds themselves are polar they are neutralized by hydrogen bonding with each other when in the hydrophobic environment.
- This gives rise to regions of the polypeptide that form regular 3D structural patterns called 'secondary structure'.
- There are two main types of secondary structure:

  - α-helices
  - β-sheet

α-Helices:

- The α-helix is a rod like structure.
- It is one peptide chain which has coiled so that adjacent residues are 1.5A apart.
- There are H-bonds between the N-H and the C=O groups of the backbone.
- These link the peptide bonds between every fourth amino acid.
- The sidechains all extend outwards from the helix.

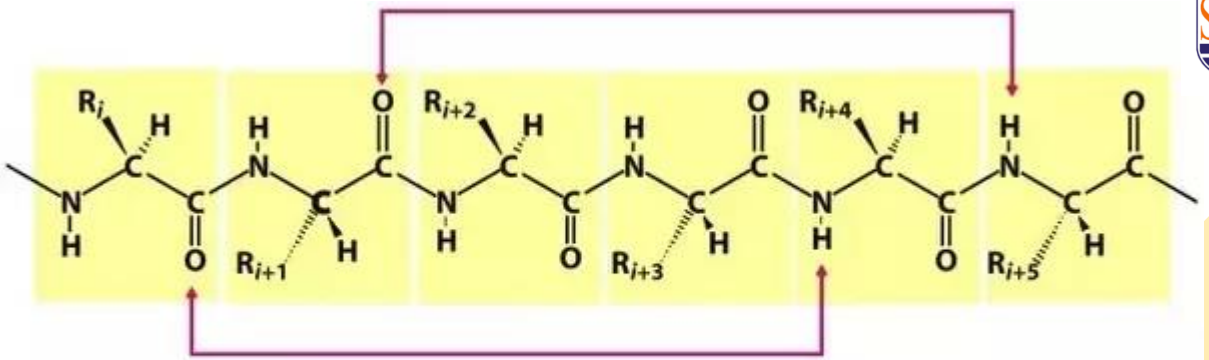


Figure 2.25
Biochemistry, Seventh Edition
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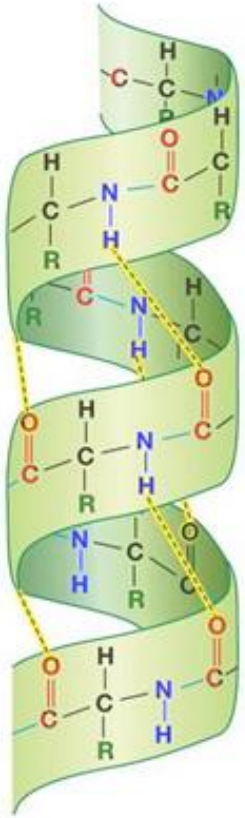
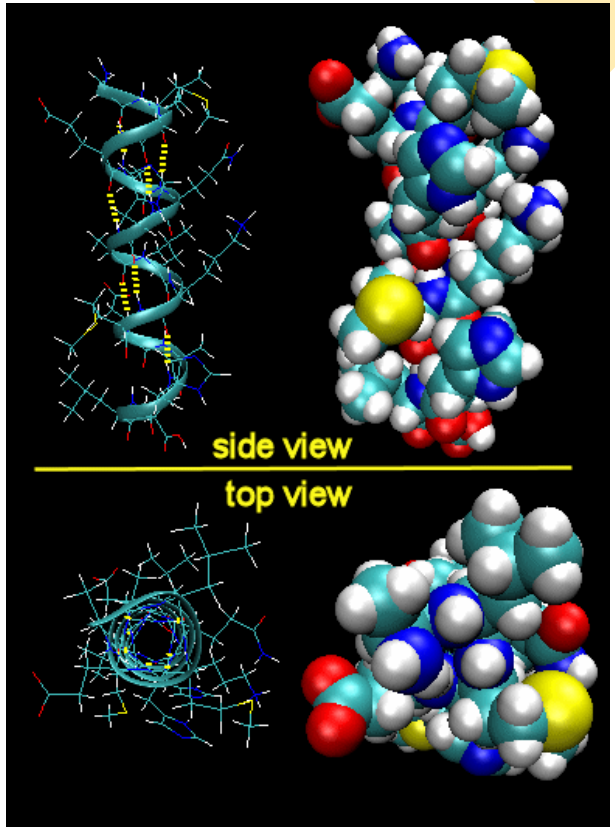
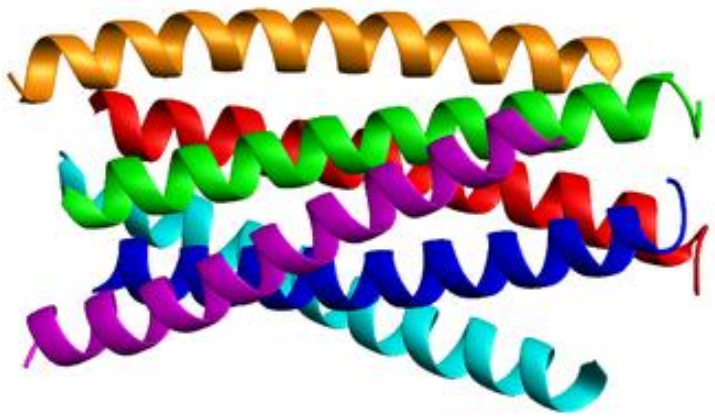




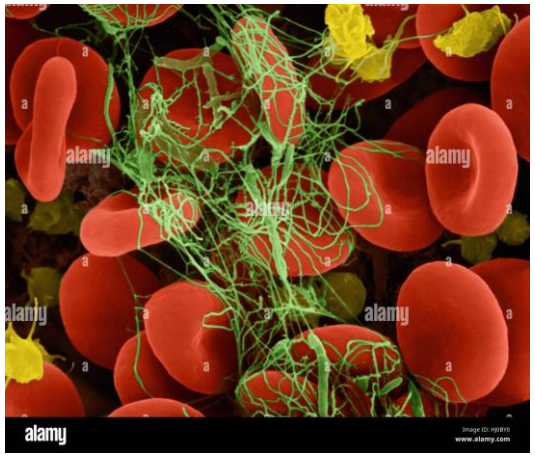
α-Helices:

- Although the helix itself is quite a rigid structure, it can be curved or kinked, which lends some flexibility to the protein structure.



- Two or more polypeptides can entwine to form very long, very stable, structures called α-helical coiled coils.

- These are often found in keratin (hair), myosin (muscle), epidermis (skin), and fibrin (blood clots).



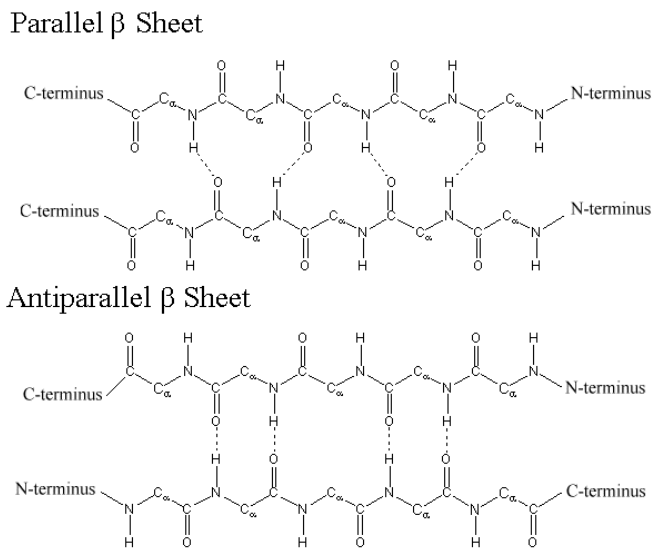
- The coiled coils serve a mechanical role for the proteins mentioned above, which are all fibrous proteins.

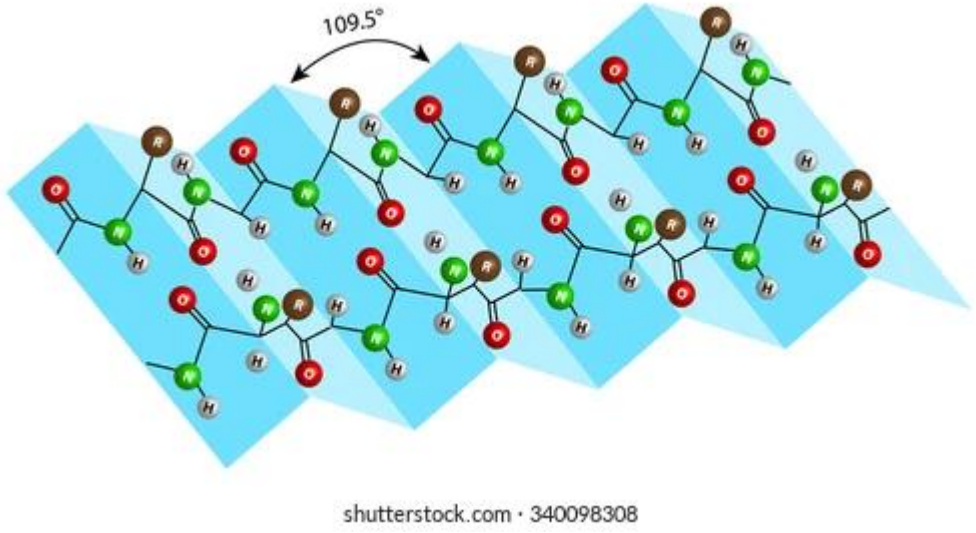
β-Sheets

- This secondary structure differs considerably from the α-helix.
- However, β-sheets are classified with α-helices since they are a regular structural element and are held together by hydrogen bonds between the -N-H and C=O groups of the peptide bond.

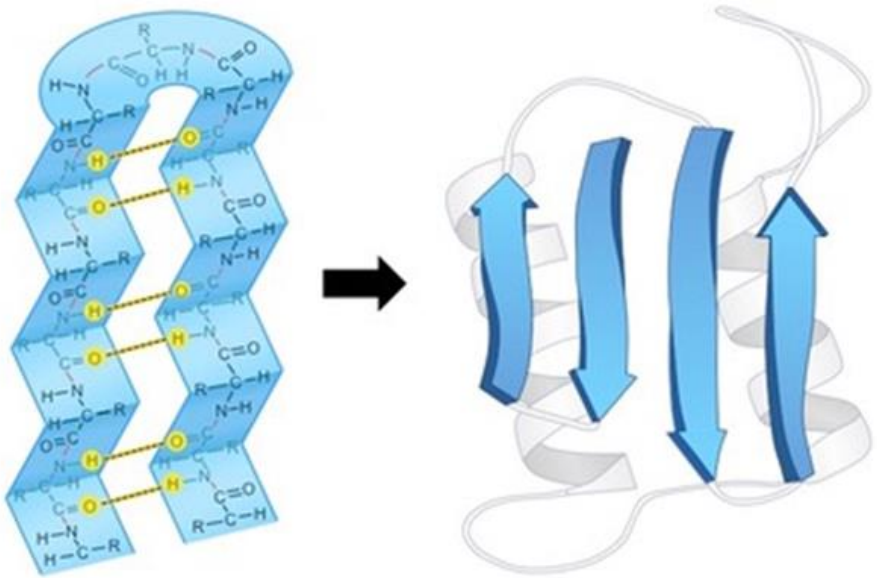
- A polypeptide in a β-sheet is called a β-strand and is almost fully extended (not coiled) so there is 3.5A between adjacent residues.

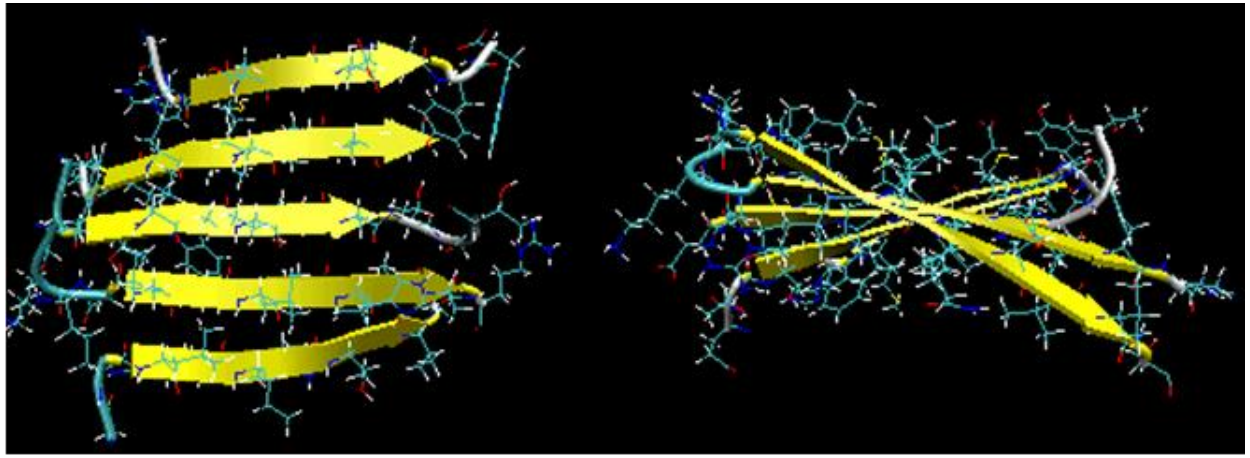
- The β-sheet is stabilized by H-bonds between the N-H and the C=O groups of different peptides or different parts of the same peptide. The β-strands in a sheet can either run in the same (parallel) or different (anti-parallel) direction.



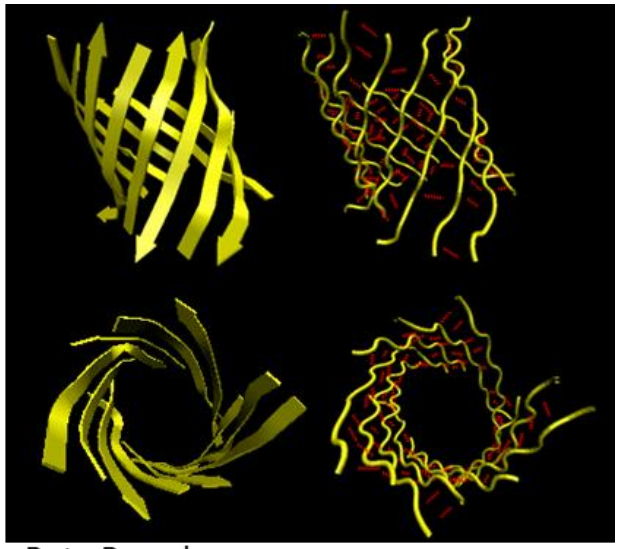


β-Sheets arrangements:





Twisted Beta Sheet/Saddle



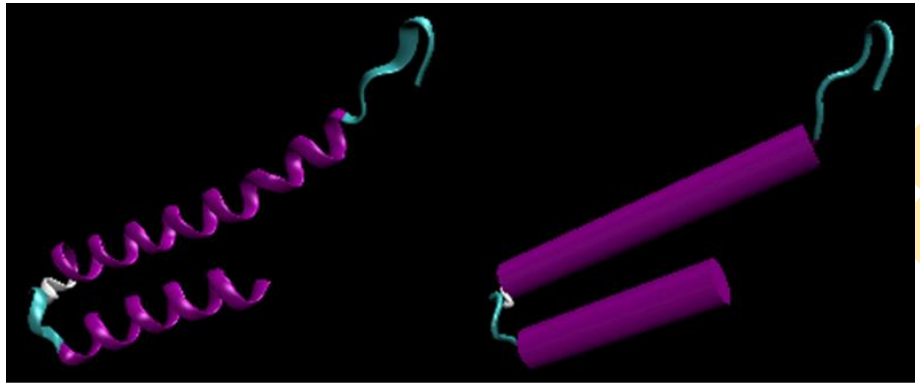
Beta Barrel

MOTIFS

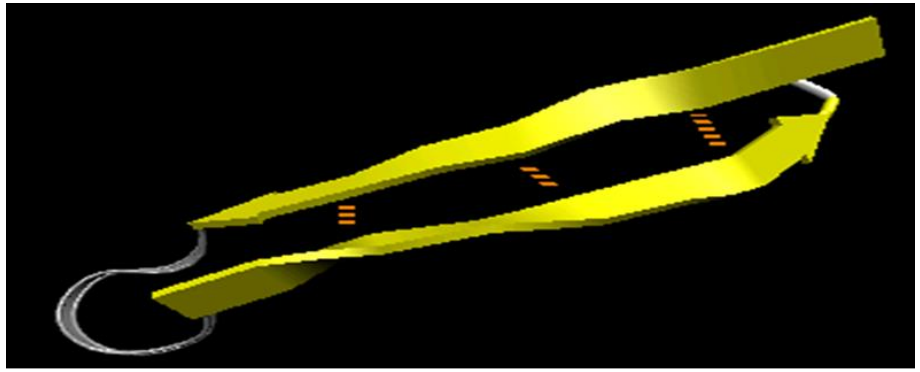
- Some simple combinations of secondary structure elements have been found to frequently occur in protein structure and are referred to as 'super-secondary structure' or motifs.

- For example, the β-hairpin motif consists of two adjacent antiparallel β-strands joined by a small loop. It is present in most antiparallel β structures both as an isolated ribbon and as part of more complex β-sheets.
- Another common super-secondary structure is the β-α-β motif, which is frequently used to connect two parallel β-strands.

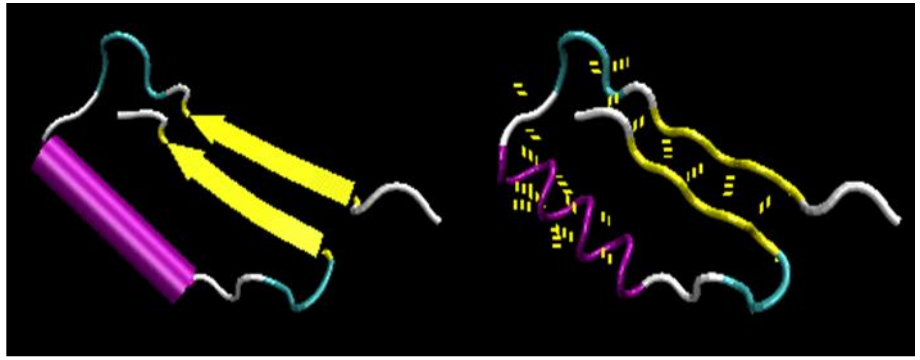
- The central α-helix connects the C-termini of the first strand to the N-termini of the second strand, packing its side chains against the β-sheet and therefore shielding the hydrophobic residues of the β-strands from the surface.



helix-loop-helix



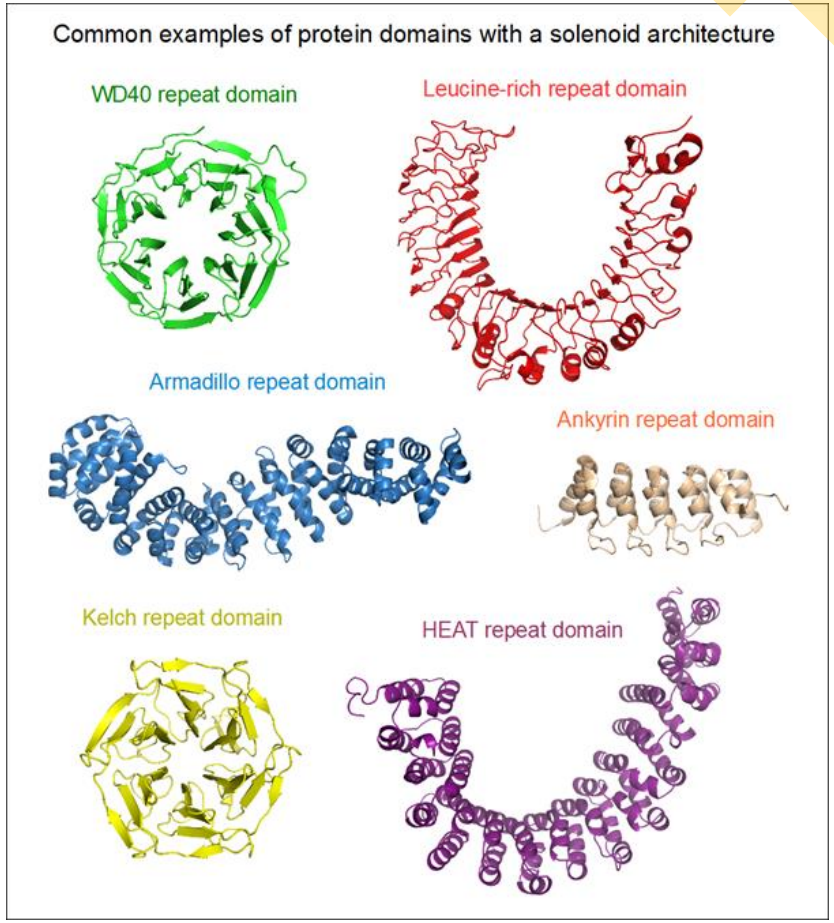
beta-hairpin, or beta-beta



beta-alpha-beta

DOMAINS:

- A protein domain is a region of the protein's polypeptide chain that is self-stabilizing and that folds independently from the rest. Each domain forms a compact folded three-dimensional structure.
- In general, domains vary in length from between about 50 amino acids up to 250 amino acids in length.
- Several motifs pack together to form compact, local, semi-independent units called domains.
- The overall 3D structure of the polypeptide chain is referred to as the protein's 'tertiary structure.



DOMAINS:

- Domains are the fundamental units of tertiary structure, each domain containing an individual hydrophobic core built from secondary structural units connected by loop regions.

- The packing of the polypeptide is usually much tighter in the interior than the exterior of the domain producing a solid-like core and a fluid-like surface.

- In fact, core residues are often conserved in a protein family, whereas the residues in loops are less conserved, unless they are involved in the protein's function.

- Protein tertiary structure can be divided into four main classes based on the secondary structural content of the domain.

DOMAINS

- The concept of the domain was first proposed in 1973 by Wetlaufer after X-ray crystallographic studies of hen lysozymelysozyme (Phillips, 1966), papain (Drenth et al., 1968) and by limited proteolysis studies of immunoglobulins (Porter, 1973; Edelman, 1973). Wetlaufer defined domains as stable units of protein structure that could fold autonomously.

- Definition: a compact structural domain that is found amongst diverse proteins is likely to fold independently within its structural environment.

- In a multidomain protein, each domain may fulfill its own function independently, or in a concerted manner with its neighbors.
- Domains can either serve as modules for building up large assemblies such as virus particles or muscle fibers or can provide specific catalytic or binding sites as found in enzymes or regulatory proteins.

DOMAINS (Classes)/ Structural classification (Tertiary structure)

- All α-domains proteins have a domain core built exclusively from α-helices.
- All β-domains proteins have a core comprising of antiparallel β-sheets, usually two sheets packed against each other.
- α+β domains proteins are a mixture of all-α and all-β motifs.
- α/β domains proteins are made from a combination of β-α-β motifs that predominantly form a parallel β-sheet surrounded by amphipathic α-helices.

STRUCTURAL CLASSIFICATION OF PROTEINS

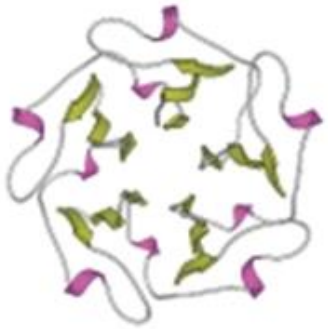
SCOP classes: 1.71 release							
All-α	All-β	α/β	α+β	Multi-domain	Membrane and cell surface	Small proteins	Other proteins
226-Folds	149-Folds	134-Folds	286-Folds	48-Folds	49-Folds	79-Folds	189-Folds
							Coiled coil (7) Low resolution (24) Peptides (116) Designed proteins (42)

DOMAINS (Classes)

α Horseshoe (1lshA02)

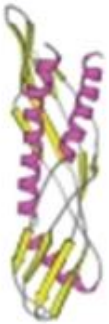
α solenoid (1pprM01)

αα barrel (1fce001)



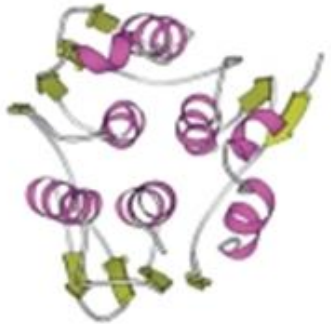
5-bladed β propeller (1tl2A00)

3-layer (βββ) sandwich (1tg7A02)



α/β super-roll (1ewfA01)

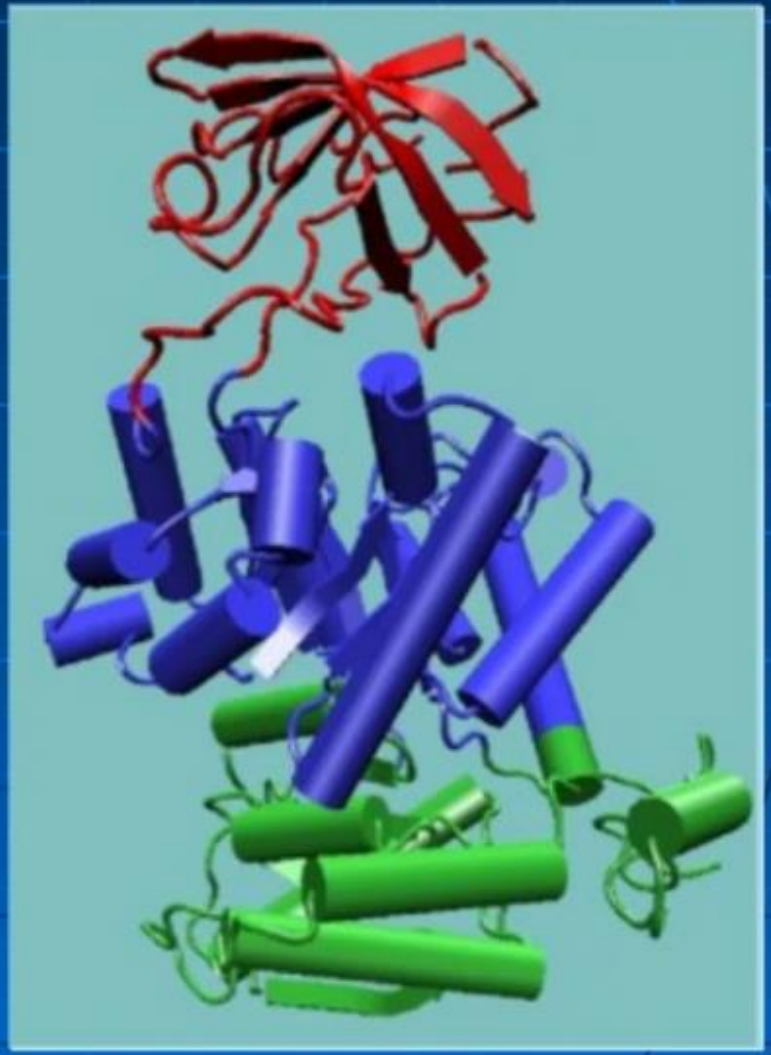
3-layer (βαβ) sandwich (1j5uA01)



α/β prism (1g6sA01)

DOMAINS

- An appropriate example is pyruvate kinase, a glycolytic enzyme that plays an important role in regulating the flux from fructose-1,6-biphosphate to pyruvate. It contains an all-β regulatory domain, an α/β-substrate binding domain and an α/β-nucleotide binding domain, connected by several polypeptide linkers (George and Heringa, 2002). Each domain in this protein occurs in diverse sets of protein families.

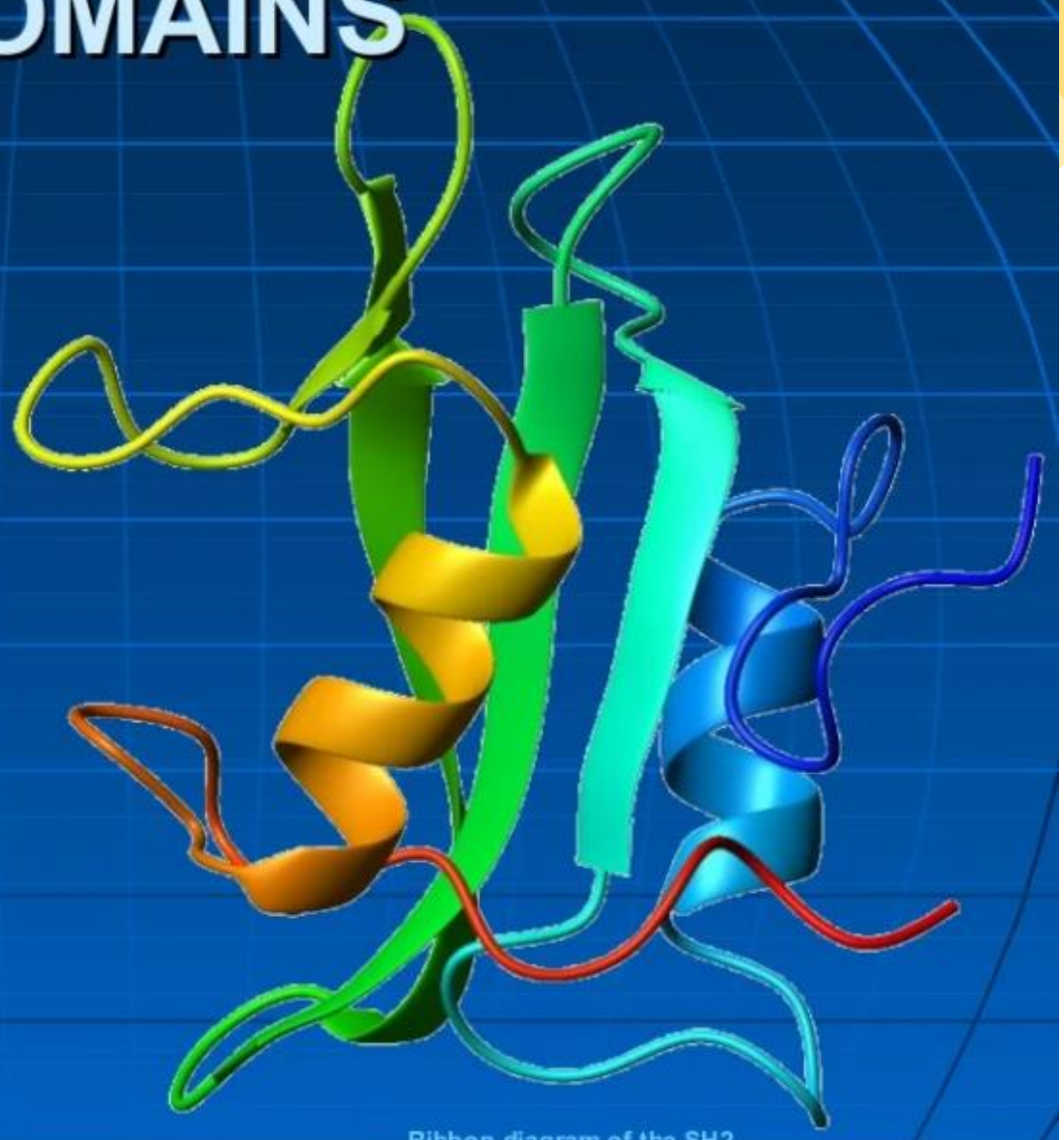


Pyruvate kinase

DOMAINS

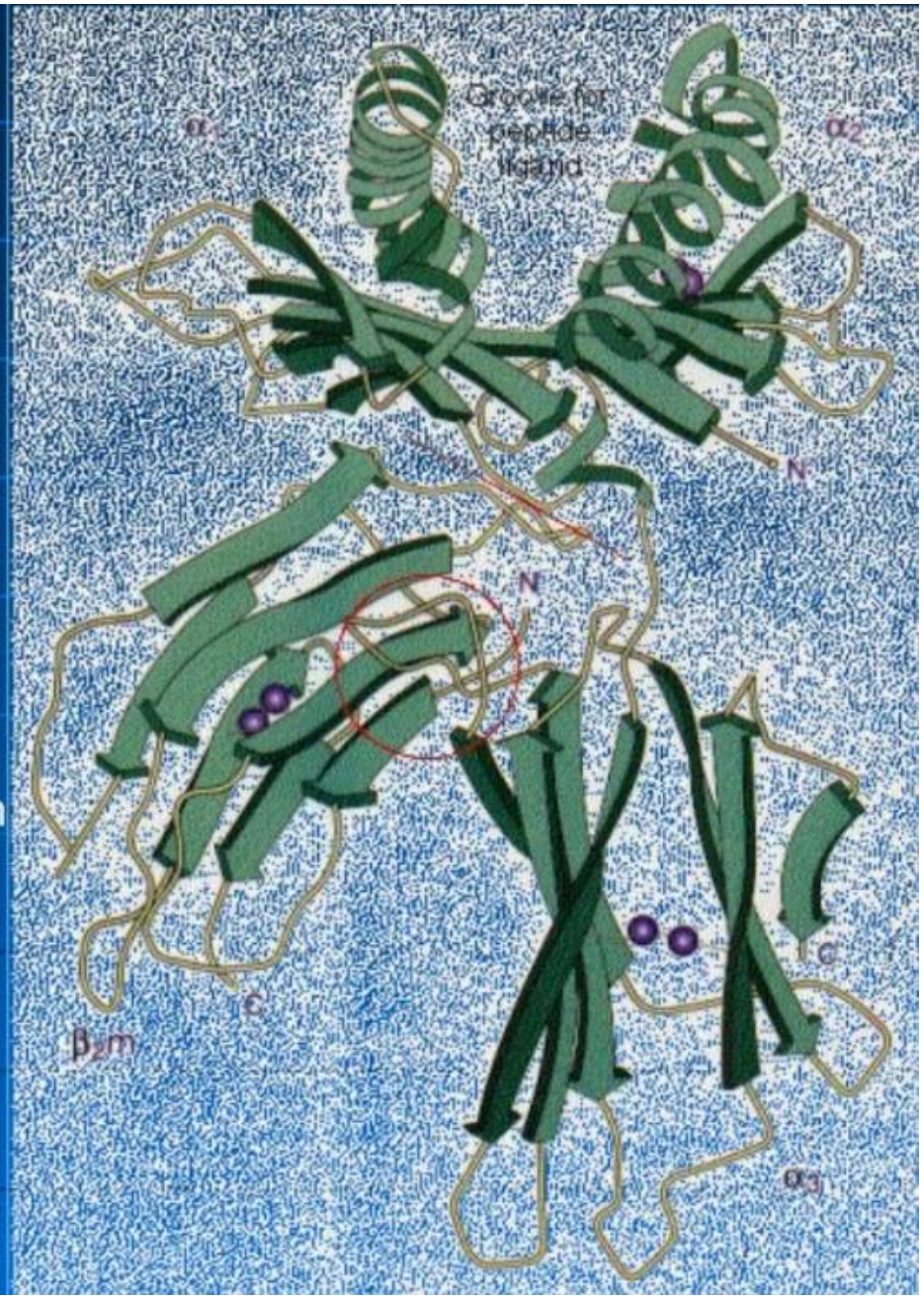
The **Src homology 2 domain** (or **SH2 domain**) is a protein domain of about 100 amino acid residues first identified as a conserved sequence region among the oncoproteins Src and Fps.
SH2 domains typically bind a phosphorylated tyrosine residue
Phosphorylation of tyrosine residues in a protein occurs during signal transduction and is carried out by tyrosine kinases. In this way, phosphorylation of a substrate by tyrosine kinases acts as a switch to trigger binding to an SH2 domain-containing protein.

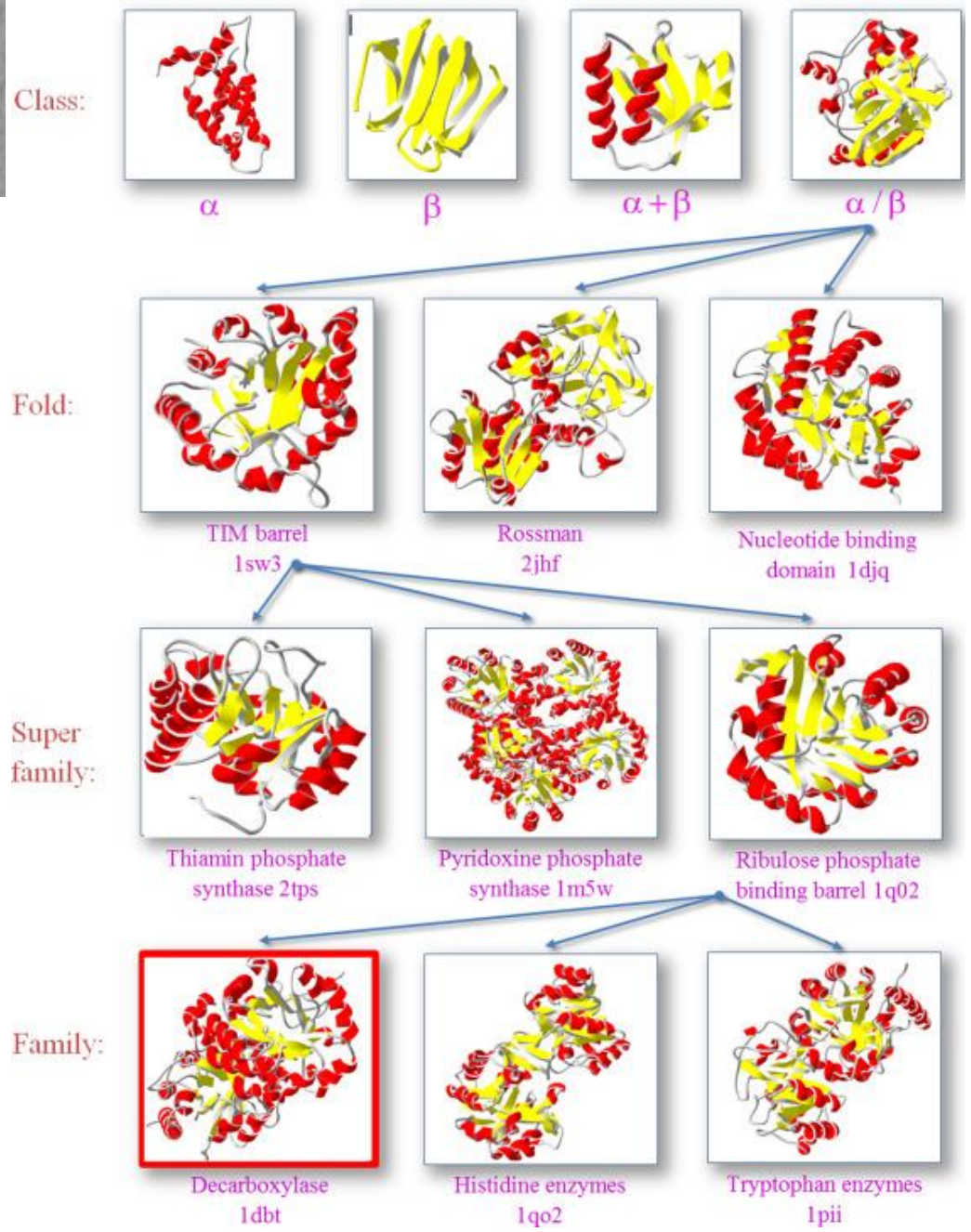
Pawson, T., Gish, G., Nash, P. (2001). SH2 domains, interaction modules and cellular wiring. Trends in Cell Biology 11(12): 504-511



Ribbon diagram of the SH2 domain from P56-Lck tyrosine kinase

This image (courtesy of P. J. Bjorkman from Nature 329:506, 1987) is a schematic representation of the extracellular portion of HLA-A2, a human class I histocompatibility molecule. It also illustrates two common examples of secondary structure: the stretches of **beta conformation** are represented by the broad green arrows (pointing N -> C terminal); regions of **alpha helix** are shown as helical ribbons. The pairs of purple spheres represent the disulfide bridges.





- The Structural Classification of Proteins: SCOP is to some a magical hierarchical classification of protein structures.
- SCOP classifies protein structures at many levels from overall class (all alpha helix, all beta strand etc.) down to protein domain and species level.
- The SCOP tree takes the protein domain as the evolutionary unit and represents evolutionary relationships between proteins such that anything classified within a superfamily is evolutionarily related.
- Whereas protein domains within the same class or fold are not necessarily evolutionarily related.

Tertiary Structure (3o/ 3D):

- Tertiary structure—the three-dimensional structure of a protein—is the next level of complexity in protein folding.
- Whereas individual amino acids in the primary sequence can interact with one another to form secondary structures such as helices and sheets, and individual amino acids from distant parts of the primary sequence can intermingle via charge-charge, hydrophobic, disulfide, or other interactions, the formation of these bonds and interactions serve to change the shape of the overall protein.
- The folding that we end up with for a given polypeptide is the tertiary structure.
- The tertiary structure of proteins deals with how the regional structures are put together in space.
- For example, the α-helices may be oriented parallel to each other or at right angles.
- The tertiary structure refers to the folding of the different segments of helices, sheets, turns, and the remainder of the protein into its native three-dimensional structure.
- Commonly, membrane proteins are anchored in the membrane by hydrophobic alpha helices that may be far removed in the primary sequence but closely apposed in the tertiary structure of the protein.

Tertiary structure bonds

	Nature of Bonding	Example
Hydrophobic interactions	Attractions between nonpolar alkyl and aromatic groups form a nonpolar center that is repelled by water	$—CH_3$ $CH_3—$ $—CH_2OH \cdots\cdots O(H)—H$
Hydrophilic Interactions	Attractions between polar or ionized R groups and water on the surface of the tertiary structure	
Salt bridges	Ionic interactions between ionized R groups of acidic and basic amino acids	$—C(=O)O^- \cdots\cdots H—N^+(H)—$
Hydrogen bonds	Occur between polar side groups of amino acids	$>C=O \cdots\cdots HO—$ $>C=O \cdots\cdots H—N(H)—$
Disulfide bonds	Strong covalent links between sulfur atoms of two cysteine amino acids	$—SH + HS— \longrightarrow —S—S—$

Hydrophobic Interactions

- Non-Polar/Hydrophobic/Aliphatic sidechains cannot hydrogen bond with water, hence the network of water is disrupted by them. This is avoided if the non-polar sidechains are removed or shielded from the water and buried in the interior of the protein. In this coiled coil structure note that the interior of the structure is occupied solely by hydrophobic residues.
- Hydrophobic interaction are a major driving force in protein folding. Hydrophobic interactions are also important in the formation of secondary structure elements such as a-helices and b-sheets. The peptide backbone is relatively hydrophillic because of the C=O and N-H groups of each peptide bond. However in the centre of the protein these tend to be in a hydrophobic environment. One way to reduce the hydrophilicity of the peptide backbone is to for hydrogen bonds between the peptide atoms. The a-helix and b-sheet are two structures which maximise the hydrogen bonding between the peptide bonds of the backbone and reduce its hydrophilicity.

Hydrogen Bonds

- When unfolded, all polar/hydrophillic sidechains can interact via H-bonds with water. When the protein folds, they must H-bond to each other and exclude much of the water. All groups capable of forming a hydrogen bond MUST, hence H-bonding in the backbone (C=O to N-H) by way of helices and sheets is an efficient way of ensuring maximum H-bonding.
- Sidechains can either accept (as in C=O) or donate (as in N-H, or O-H) an H-bond.
- The capacity of proteins to form hydrogen bonds is an important determinant of protein stability. Hydrogen bonds can be between backbone groups, as in helices and sheets; between side chains, such as serine or threonine O-H groups and carbonyl carbons of side chains (-C=O); and between backbone groups and side chain groups.

Ion Pairs

- When amino acid sidechains of opposite charge are in close proximity, they can form an ion pair (also called a salt bridge). But they are also capable of hydrogen bonding and hence, are usually found on the surface of the protein. If they can form a salt bridge, they will usually be buried. Since charge is affected by pH, so is the formation and the breakage of these ion pairs.
- Salt bridges also increase the stability of the tertiary structure.

  NOTE: **Hydrophobic interactions, hydrogen bonds and salt bridges are all non-covalent interactions**
- These are all relatively weak interactions but the large number in a protein combine to give the overall stability of the structure.

Disulphide Bonds

- If two cysteine sidechains are close to one another and the local environment is conducive to oxidation,then they can form a disulphide bond/bridge as shown. The residues can either be in the same peptide chain (forming a loop) or in different chains.
- As Disulphide bonds are covalent linkages they add considerably to the stability of proteins in which they occur. However they mainly occur in secreted proteins and in the parts of membrane proteins which face the outside.

Note that disuphide bridges only form after the protein has folded into its tertiary structure. This folding is driven by the combined energy of all of the weak interactions described in the previous section above.

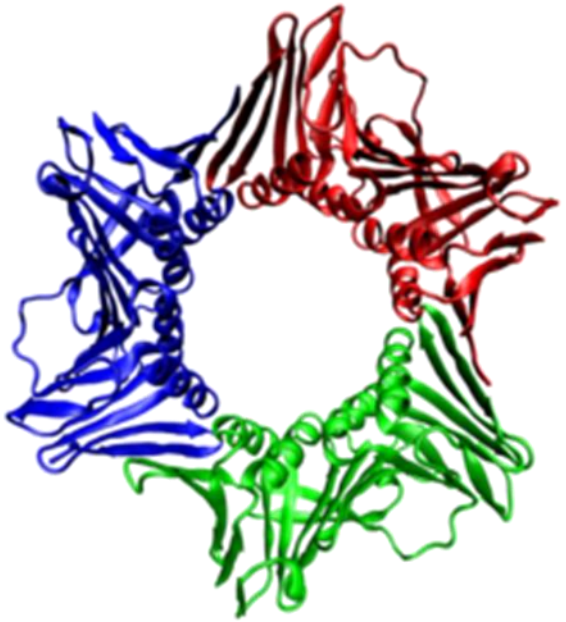
Cysteine side chains in cytosolic proteins are usually found as -SH groups as the cytosolic environmemt is sufficiently reducing to prevent disulphides from forming.

Non-Covalent Bonds

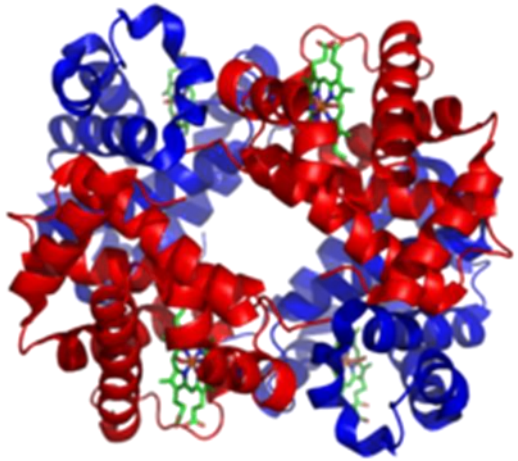
- The tertiary structure of proteins is maintained by a host of weak, non-covalent interactions:
  - Hydrophobic Interactions
  - Hydrogen Bonds
  - Salt Bridges
- Protein stability is a function of many weak interactions. If these interactions are maintained the protein keeps its tertiary ("native") stucture. However once the bonds maintaining the structure begin to be disrupted the tertiary structure is destroyed and the protein is said to be "denatured"

Quaternary Structure

- Many proteins contain more than one polypeptide (or subunit) which associate in a specific way to make large assemblies out of smaller replicable units.
- Also, enzymes with many subunits provides a way to regulate them.
- Subunit interactions-contact regions are complementary surfaces held together by
    - hydrophobic bonds
    - hydrogen bonds
    - van der waals interactions
    - salt bonds
    - occasionally di-sulfide bonds
- Symmetry-in majority of multi subunit proteins the subunits are arranged with geometrically equivalent positions.
- Examples of proteins with quaternary structure include hemoglobin, DNA polymerase, ribosomes, antibodies, and ion channels.



Proliferating cell nuclear antigen (PCNA)



Hemoglobin. α and β subunits are in red and blue

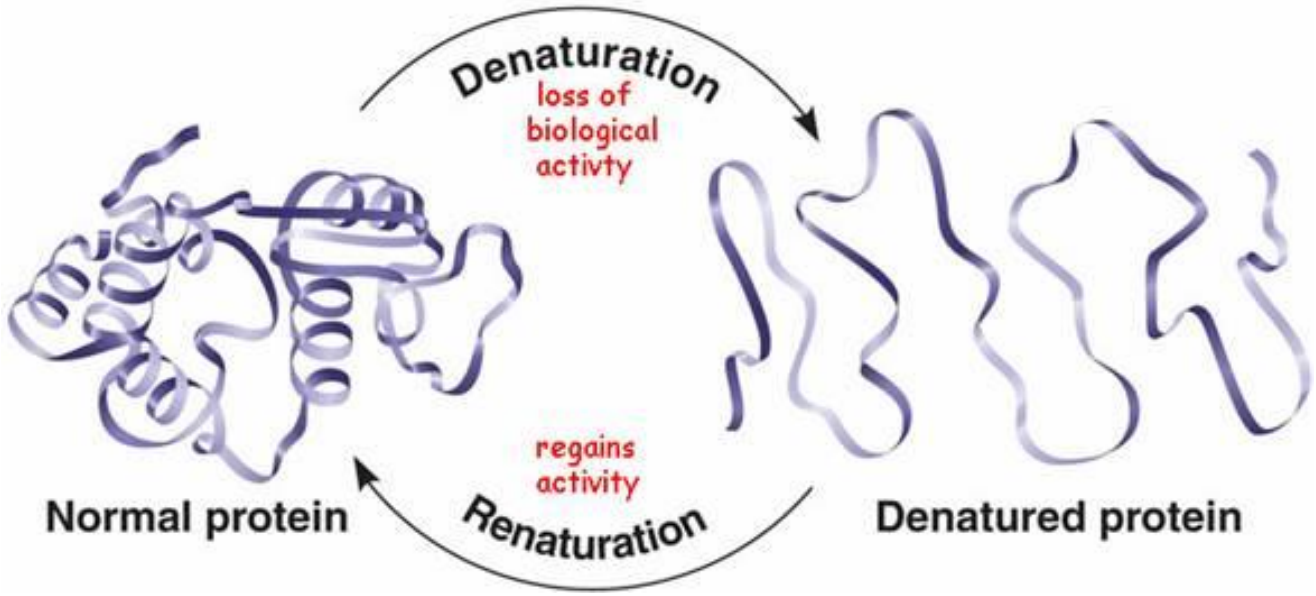
Subunits & Quaternary Structure

- Proteins containing more than one polypeptide chain exhibit an additional level of structural organisation.
- In this case, each polypeptide chain is a "subunit". Quaternary (4o) structure describes the way the subunits are arranged together and the nature of their contacts.
- Subunits associate by non-covalent interactions similar to those involved in tertiary structure:
  - **Hydrophobic Interactions**
  - **Hydrogen bonds**
  - **Salt Bridges / Ion Pairs**

Denaturation

- The process that causes a protein to lose its shape is known as denaturation.

- Changes in pH, Increased Temperature, Exposure to UV light/radiation (dissociation of H bonds), Protonation amino acid residues, High salt concentrations are the main factors that cause a protein to denature.



Factors Which Can Denature A Protein

- Heat: increase atomic vibrations, ruins interactions, need heat of >50°C to denature most proteins.
- pH: breaks the salt bridges.
- Organic Solvents: disrupt hydrophobic interactions
- Detergents: hydrophobic part pushes into protein core
- When a protein is denatured, it loses it's 3D shape. When a protein is denatured, it loses it's biological activity. Therefore...

A PROTEINS TERTIARY STRUCTURE IS ESSENTIAL TO IT'S BIOLOGICAL ACTIVITY

Videos

- Part 1: Protein Phosphorylation in Biology-video
  - https://youtu.be/CBr1cqCVjsk
- Part 2: Architecture of a Protein Kinase
  - https://youtu.be/TQ2C-5P-I5E

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Thank you